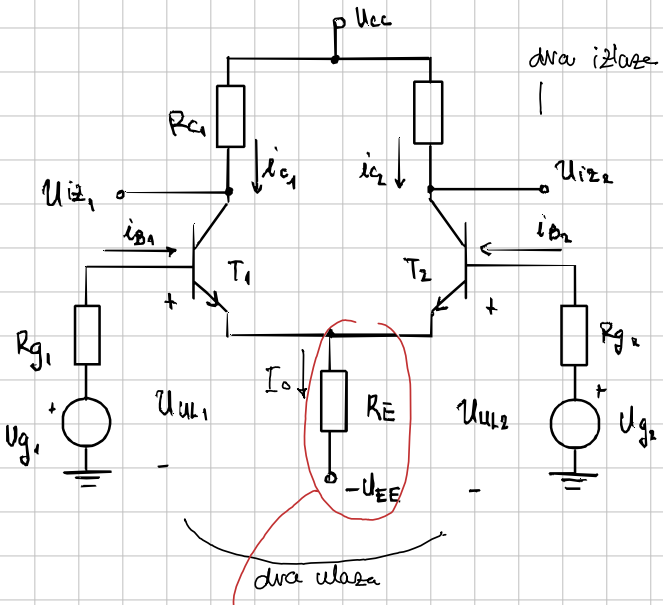
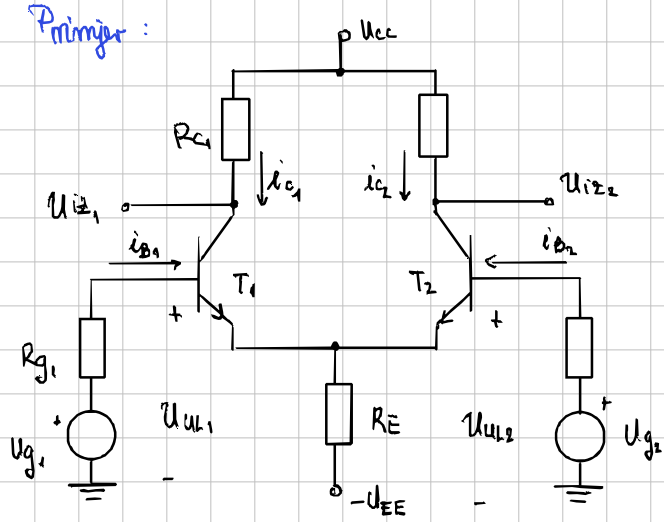
Bipolarno diferencijsko pojačalo s emiterskim otpornikom



- ako koristimo samo 1 izlaz
  → asimetrični izlaz
- razlika $U_{iz} = U_{iz2} - U_{iz1}$
  → diferencijski ili simetrični izlaz

istosmerno pojačalo

Primjer:



ako je zadovoljena potpuna simetrija
SRT → $Q_1 = Q_2$

$i_{C1} = i_{C2}$, $U_{CE1} = U_{CE2}$, sve jednako

$U_{CC} = U_{EE} = 15V$

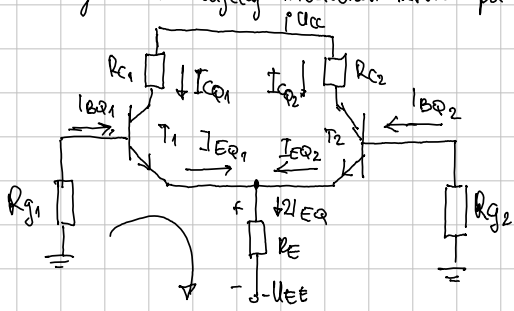
$R_{g1} = R_{g2} = 500\Omega$

$R_{C1} = R_{C2} = 1{,}5k\Omega$

$R_E = 4{,}5k\Omega$, $\beta = 100$, $U_{\gamma} = 0{,}7V$

STATIKA: gasimo izmjenične

→ gledamo utjecaj statičkih izvora pa zato gasimo dinamičke



$I_{BQ1} = I_{BQ2} = I_{BQ}$ ← jer nijedi potpuna simetrija

$I_{CQ1} = I_{CQ2} = I_{CQ}$

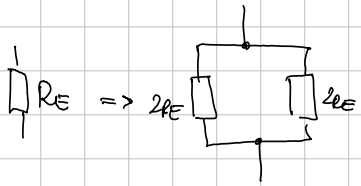
$I_{EQ1} = I_{EQ2} = I_{EQ}$

$I_{EQ} = (1+\beta) I_{BQ}$

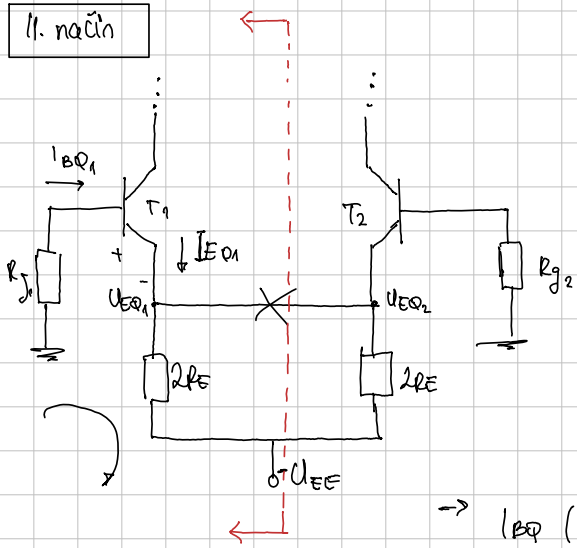
<u>ulazni krug T_1:</u> $I_{BQ} \cdot R_{g1} + U_{BEQ} + 2 I_{EQ} \cdot R_E - U_{EE} = 0$

$I_{BQ}(R_g + 2(1+\beta) \cdot R_E) = U_{EE} - U_{BEQ}$

$$I_{BQ} = \frac{U_{EE} - U_{BEQ}}{R_g + 2(1+\beta) \cdot R_E} \overset{\text{uvrsti}}{=} \dots = 15{,}7\mu A$$



II. način



$U_{EQ_1} = U_{EQ_2} \longrightarrow$ te dvije točke su na istom potencijalu $\rightarrow$ nema struje

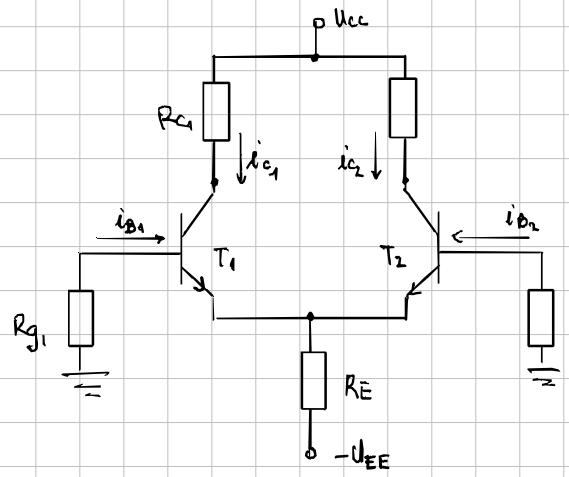
↳ možemo promatrati samo jednu polovinu pojačala

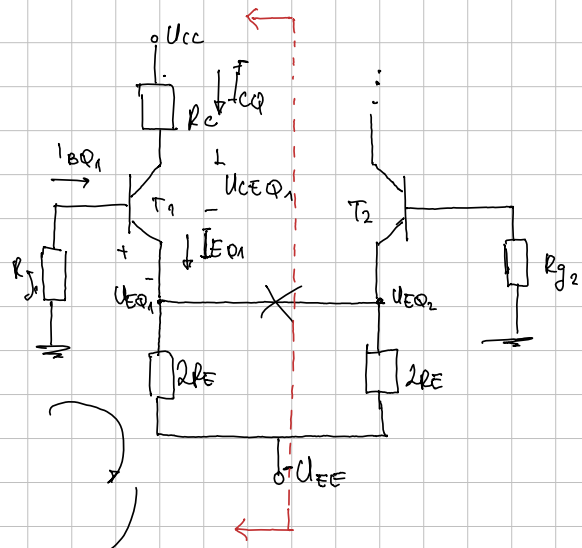
$$R_{g_1} \cdot I_{BQ_1} + U_{BEQ_1} + I_{EQ_1} \cdot 2 \cdot R_E - U_{EE} = 0$$

$$I_{EQ} = (1+\beta)\, I_{BQ}$$

$$\rightarrow \quad I_{BQ}\left(R_g + 2(1+\beta)\cdot R_E\right) = U_{EE} - U_{BEQ}$$

$$I_{BQ} = \frac{U_{EE} - U_{BEQ}}{R_g + 2(1+\beta)\cdot R_E}$$ dobijemo isto

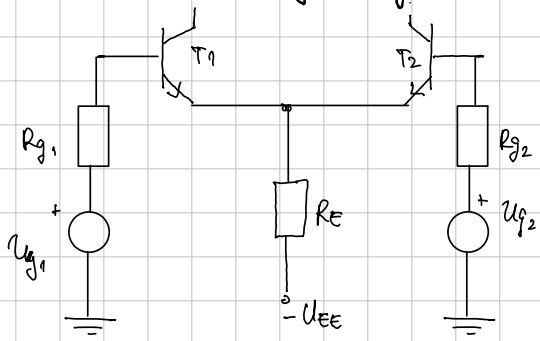




$$I_{CQ} = \beta \cdot I_{BQ} = \dots = 1{,}57\,mA$$

$$U_{CEQ} = U_{CEQ_1} = U_{CEQ_2} = U_{CC} - (-U_{EE}) - I_{CQ}\cdot(R_C + 2R_E) = \dots = 13{,}5\,V > U_{\gamma}$$

možemo značajke signala izvući iz signala koje dovodimo na ulaz



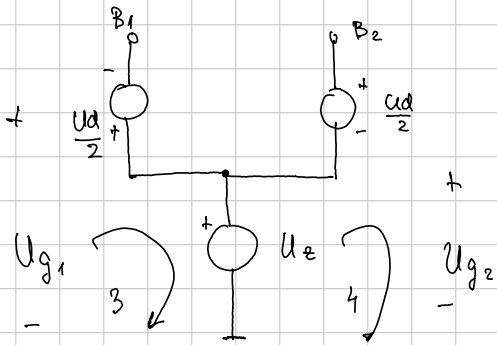
Ulazni → U_{g1} i U_{g2}

(1) ZAJEDNIČKI SIGNAL: → $U_z = \frac{U_{g1}+U_{g2}}{2}$

(2) DIFERENC. SIGNAL: $U_d = U_{g2} - U_{g2}$

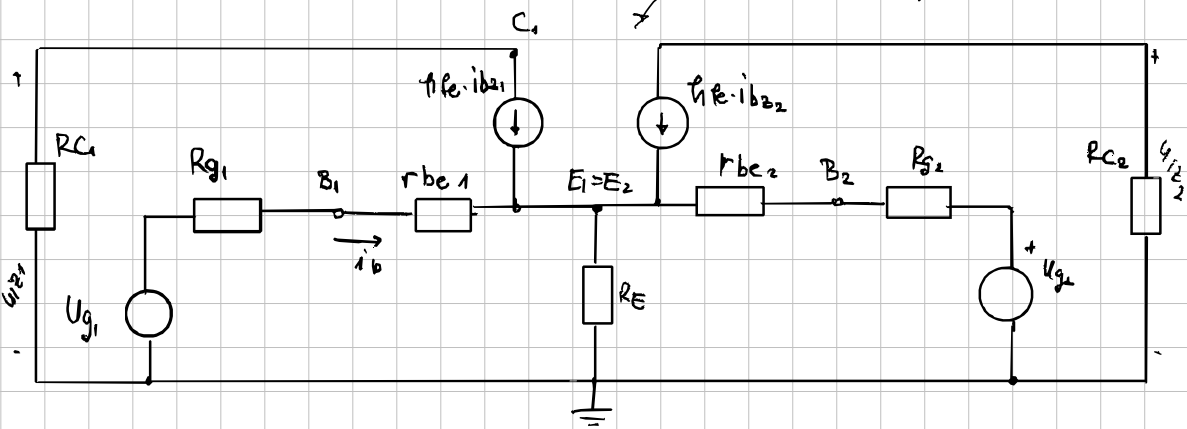
$$\boxed{\begin{aligned} U_{g1} &= U_z - \frac{U_d}{2} \\ U_{g2} &= U_z + \frac{U_d}{2} \end{aligned}}$$

1. korak 2. korak superpozicija



DINAMIKA::

linearizacija sklopa

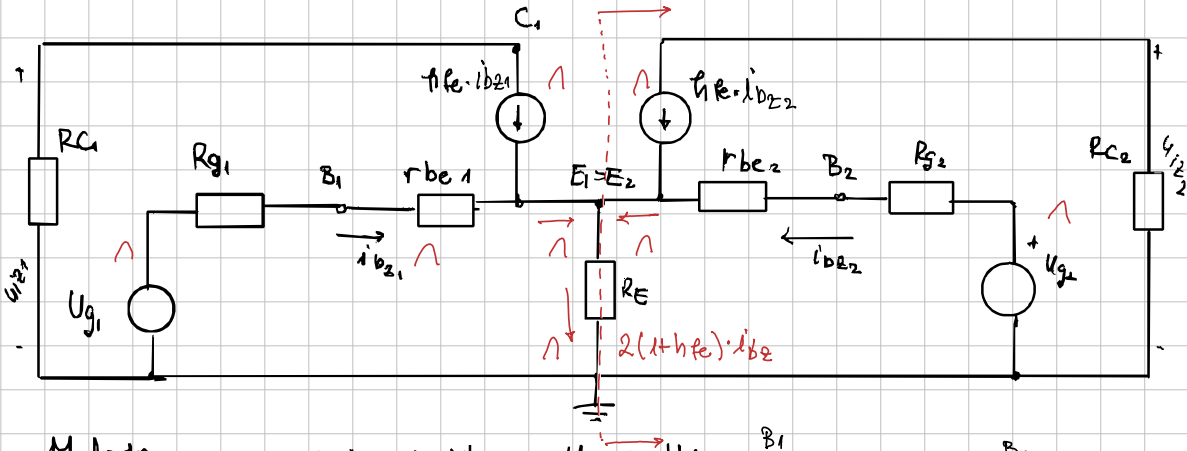


Primjer 1.2) Q iz 1.1

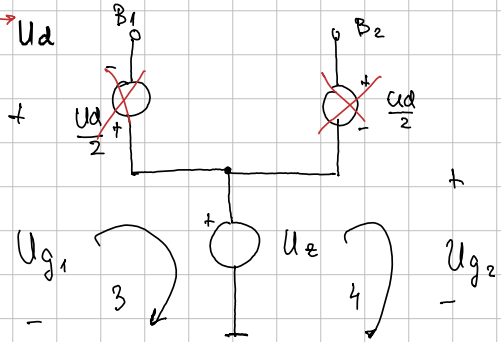
Dinamički parametri

$$r_{be} = r_{be_1} = r_{be_2} = \frac{U_T}{I_{BQ}} = \dots = 1.59\ k\Omega$$

ZAJEDNIČKI SIGNAL:



Metodom superpozicije odvojiti na U_z i U_d



Zajednički signal: U_z

- na ulaze je dovedan samo U_z

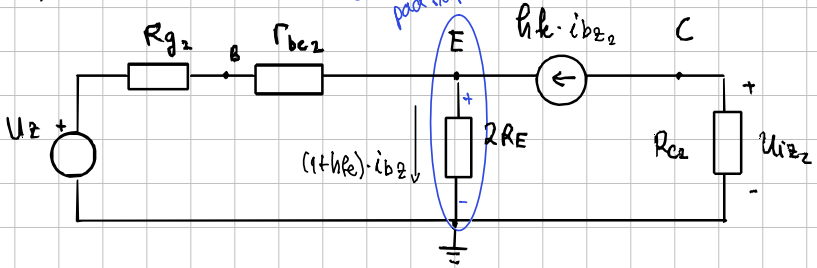
↳ kratko spajamo $\frac{U_d}{2}$

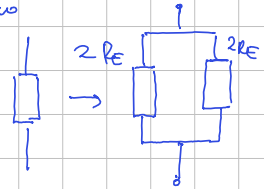
- imamo simetriju pa možemo gledati opet

- U_{g_1} i U_{g_2} dolaze istofazno → sve struje su iste faze, tj. simetrične

polovica:

da bi bio isti pad napona kao na velikoj shemi → $R_E \to 2R_E$ → ili, shvatimo R_E kao





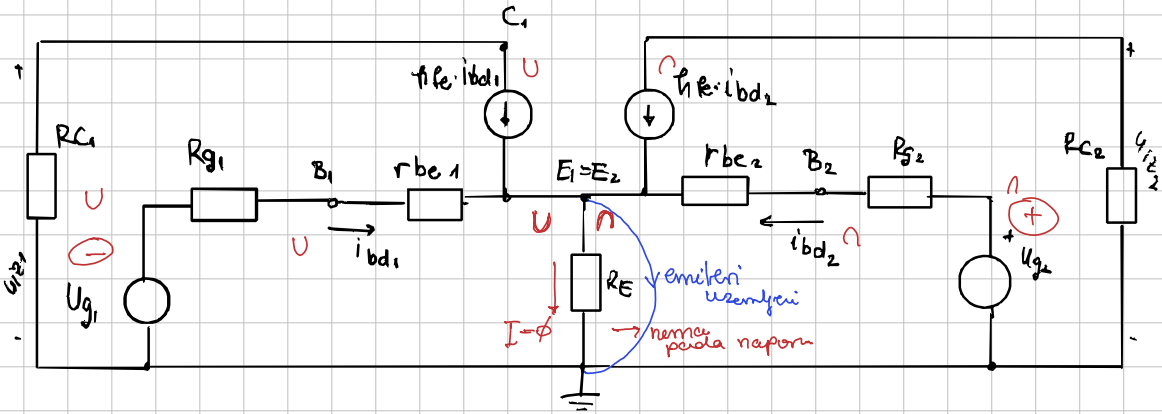
$$A_{VZ} = \frac{u_{iz_2}}{U_z} = \frac{-h_{fe}\cdot \cancel{i_{b_2}}\cdot R_{C_2}}{(1+h_{fe})\cdot \cancel{i_{b_2}}\cdot (R_{g_2}+r_{be_2}+2R_E)} = \frac{-h_{fe}\cdot R_{C_2}}{(1+h_{fe})(R_{g_2}+r_{be_2}+2R_E)} = \dots \approx -0.165$$

$(1+h_{fe})\cdot 2R_E$

degeneracija

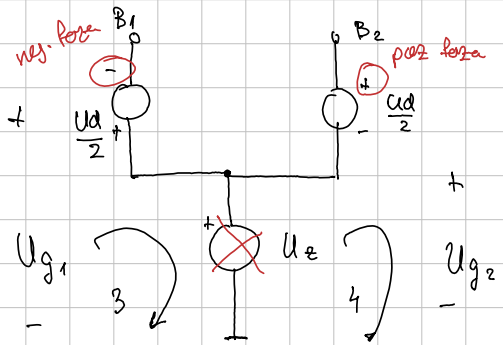
gušenje

DIFERENCIJSKI SIGNAL:

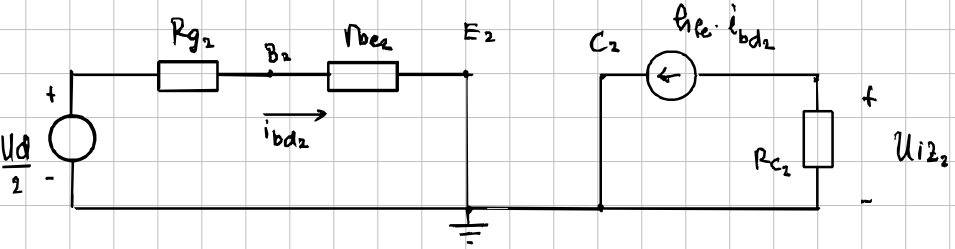


kratko spojamo Uz → na ulaze dovodimo Ud

↳ uslijed simetrije
⟹ faze su suprotne, amplitude su jednake
⇓
$I_E = 0$



- gasimo Uz
- dolaze u protufazi



$$A_{vd_2} = \frac{u_{iz_2}}{u_d} = \frac{u_{iz}}{2\cdot\frac{u_d}{2}} = \frac{-h_{fe}\cdot \cancel{i_{bd_2}}\cdot R_{C_2}}{2\cdot \cancel{i_{bd_2}}\cdot(R_{g_2}+r_{be_2})} = \frac{-1}{2}\cdot\frac{h_{fe}\cdot R_{C_2}}{R_{g_2}+r_{be_2}} = \dots = -35.9$$

+ u odnosu na diferencijski, ne ulazni

pojačavaje bez degeneracije (nema R_E)

FAKTOR POTISKIVANJA

- opisuje kvaliteta pojačala

$$\rho = \left| \frac{A_{vd}}{A_{vz}} \right| = \dots = \left| \frac{-35.9}{-0.165} \right| = \underline{\underline{218}}$$

velik kod dobrih

$$\rho_{dB} = 20 \log \rho = \dots = \underline{46{,}8\ dB} \rightarrow$$ veći ≈ kvalitetnije

e.g. common mode rejection ratio (CMRR)

↳ faktor potiskivanja ZAJEDNIČKOG SIGNALA

↳ jer to je smetnja i njen želimo što više potisnuti